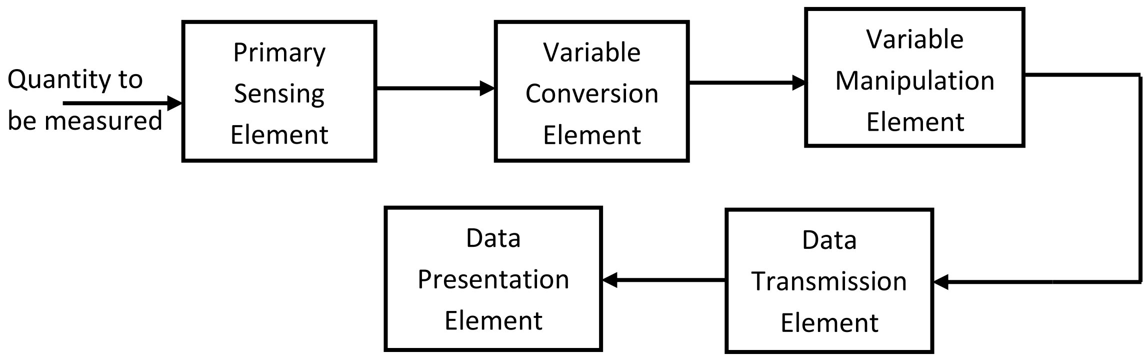
Chapter 1
Introduction to Instrumentation System

Instrumentation system

It consists of different components which together are used to perform a measurement and record the results.

Generalized block Diagram of Instrumentation System



Primary Sensing Element

The quantity under measurement makes its first contact with primary sensing element called as transducer. A transducer is a device which converts a physical (non electrical) quantity into an electrical quantity. e.g. Microphone, LDR etc.

Variable Conversion Element

The output of the primary sensing element may be electrical signal of any forms. It may be voltage or current or other electrical parameters. Sometimes this input is not suited to the system. For digital system, this analog signal will have to be converted into digital form. For example: ADC (Analog to Digital Converter)

Variable Manipulation Element

The function of this element is to manipulate the signal present to it preserving the original nature of the signal. Manipulation here means only a change in numerical values of the signal. e.g. Electronic Amplifier which accepts a small voltage signal and produces an output signal of greater magnitude.

Data Transmission Element

When the elements of an instrument are actually physically separated, it becomes necessary to transmit data from one point to another. The element that performs this task is called data transmission lines. For example, space crafts are physically separated from the earth station where the control stations guiding their movements are located. Therefore, control signals are sent from these stations to space craft by telemetry system using radio signals.

Data Presentation Element
The information about the quantity being measured should be displayed appropriately for monitoring, control and analysis purpose. In this case, data to be monitored needs visual devices. e.g Ammeter, voltmeter, LCD Display etc.

Signal and its types in Instrumentation

Signal

- Something that posses information
- It can be represented as a function of one or more independent variables.
- For example: Speech signal can be represented by acoustic pressure as a function of time.
- A picture can be represented by brightness as a function of two spatial variables (co-ordinates)

Types of signals:

1) Continuous time and Discrete time signal

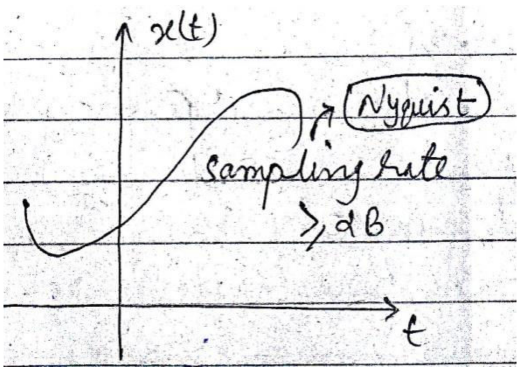


Fig: continuous Time signal

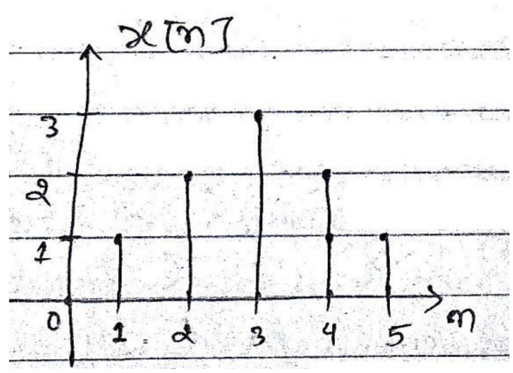
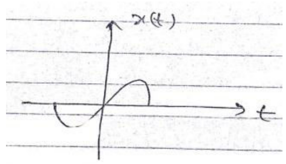


Fig: discrete time signal

In the continuous time signal, the independent variable (time) is continuous, thus the signal is specified for every values of the independent variable. It is denoted by x(t), g(t), y(t).
The discrete time signal is defined only for discrete values of time. It is represented by x[n], g[n], y[n].

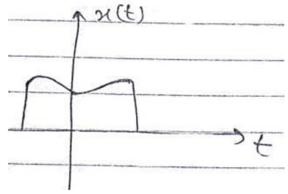
2) Deterministic and Random signal
Signal about which there is no uncertainty with respect to its value at any time and can be represented by explicit mathematical expressions is called deterministic signal. e. g $x(t)=5 \cos 10t$
Signal about which there is uncertainty before its actual occurrence and therefore cannot be expressed in the form of exact mathematical expression is called random signal. e. g Noise signal which can be analyzed by probability theory and statistical technique.

3) Even and odd Signal
A signal x(t) or x[n] is called even (symmetric) signal if it is identical to its time-reversed counterpart i.e. $x(-t)= x(t)$, $x[-n] = x[n]$



e.g. cosine wave series

A signal x(t) or x[n] is called odd signal if it isn't identical to its time-reversed counterpart i.e.
x(-t) =- x(t); x[-n] = - x[n] .e.g sine wave series



4) Energy and power signals
A signal with finite energy is called energy signal. A signal is called energy signal if and only if $0<E_x<\infty$

Where $E_x = \int_{-\infty}^{\infty} x^2(t)\, dt$ is the energy of the signal x(t)
A signal with finite power is called power signal . A signal is called a power signal if and only if $0<P_x<\infty$

Where $P_x = \lim_{T \to \infty} \int_{-T/2}^{T/2} x^2(t)\, dt$ is the power of x(t).

Note: Most periodic signals and random signals are power signals.
Most aperiodic signals and deterministic signals are energy signals

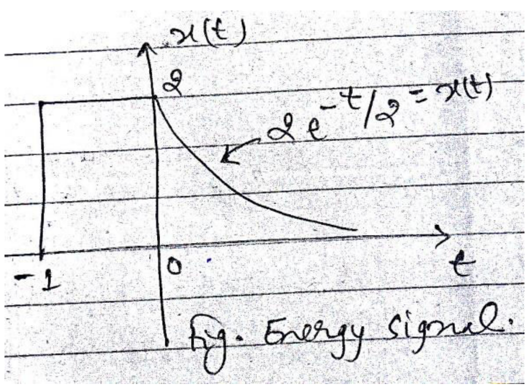


Fig: Energy signal

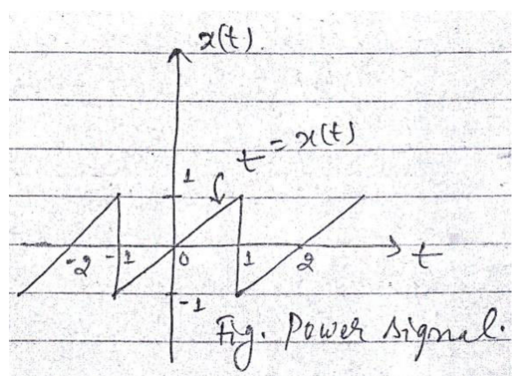


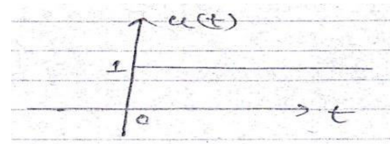
Fig: Power signal

5) Harmonic Signals
A periodic signal defined for $-\infty<t<\infty$ and expressed in terms of sinusoidal function
x(t) = $A\cos(2\pi f t + \theta)$ is called harmonic signal.

6) Unit step signal

$u(t) = 1 \quad \text{for } t \geq 0$

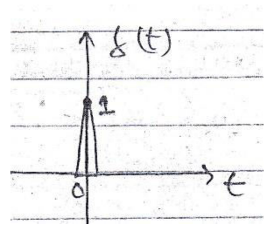
$\quad\quad\;\; 0 \quad \text{otherwise}$



7) Unit Impulse signal

$\delta(t) = 1 \quad \text{for } t = 0$

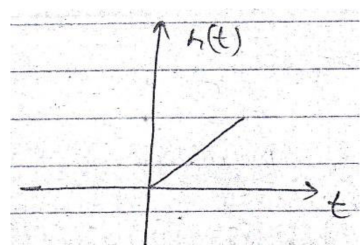
$\quad\quad\;\; 0 \quad \text{otherwise}$



8) Ramp signal

$r(t) = t \quad \text{for } t \geq 0$

$\quad\quad\;\; 0 \quad \text{otherwise}$



Chapter 2

Signal Measurement

Measurement

- It is a process by which one can convert a physical quantity to meaningful number.
- Property of an object/ system under consideration is compared to an accepted standard unit
- It is a means of describing a natural phenomenon in quantitative term.

Unit

The standard measure of each kind of physical quantity is the Unit. Following are the various type of units.

i. Fundamental (Primary) Unit
   Measures of physical quantities in length mass and time.
ii. Auxiliary Unit
   Measures of physical quantities in thermal, electrical and illumination disciplines.
iii. Derived Unit
   All other units which can be expressed in terms of the fundamental units are called derived units. For example, the derived unit for area is square meter (m^2) in SI.

Standards of measurements

A standard of measurement is a physical representation of a unit of measurement. For example, the fundamental unit of mass in international system (SI) is kilogram, defined as the mass of a cubic decimeter of water as its temperature of maximum density of $4°C$.

Types of standards of measurements:

Standards of measurement are of four types according to their function and application. They are as follows.

i. International standards
   The international standards are defined on the basis of international agreement. They are periodically evaluated and checked by absolute measurement in terms of fundamental units. The international standards are maintained at the international Bureau of weights and measures and are not available to the ordinary user of measuring instrument for purpose of comparison or calibration.

ii. Primary Standards
   The primary (basic) standards are maintained by national standards laboratories in different parts of the world. The primary standards, which represent the fundamental units and some of the derived electrical and mechanical units are independently calibrated by absolute measurements at

each of the national laboratories. One of the main functions of the primary standards is the verification and calibration of secondary standards.

iii. Secondary Standards

Secondary standards are the basic reference standards used in industrial measurement laboratories. These standards are maintained by the particular involved industry and are checked locally against other reference standards in the area. The responsibility for maintenance and calibration of secondary standards lies entirely with the industrial laboratory itself. Secondary standards are generally sent to the national standards laboratory on a periodic basis for calibration and comparison against the primary standards. They are then returned to the industrial user with a artification of their measured value in terms of the primary standard.

iv. Working Standards

Working standards are the principle tools of a measurement laboratory. They are used to check and calibrate general laboratory instrument for accuracy and performance or the perform comparison measurements in industrial applications.

v. IEEE Standards

This standard is maintained by the Institute of Electrical and Electronics Engineers, IEEE an engineering society headquartered in New York City.

These standards are not physical items that are available for comparison and checking of secondary standards but are standard procedures, nomenclature, definitions etc. It gives the standard test method for testing and evaluating various electronic systems and components. For example, there is a standard method for testing and evaluating attenuators.

Measuring Instruments

It is a device for determining the magnitude of a physical quantity being measured.

Measurement is the process by which one can convert physical parameters to numerical value, the methods of measurement can be direct and indirect. Measurement work employs a number of terms which are defined as:

i. *Accuracy*

It is the closeness with which an instrument reading approaches the true value of the variable being measured.

ii. *Precision*

It refers to the measure of the degree to which successive measurement differ from one another.

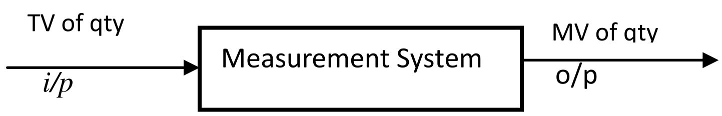
iii. *Sensitivity*

It is the ratio of output signal or response of the instrument to a change of input or measured variable.

iv. *Resolution*

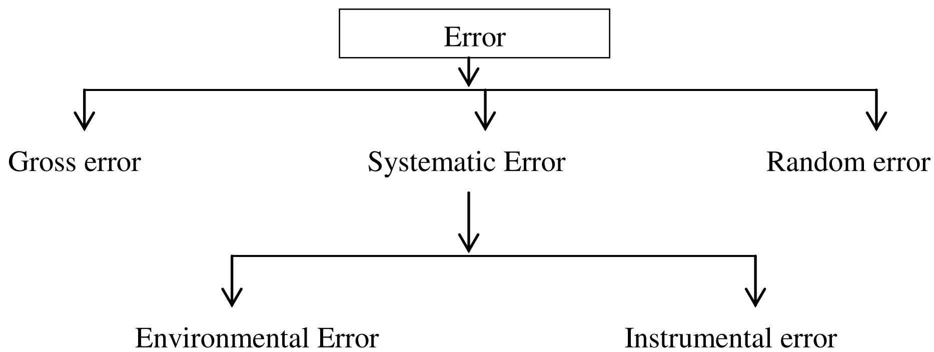
It is the smallest change in measurement value to which the instrument will respond.

v. *Error*



It is the deviation from the true value of the measured variable. Errors may come from different sources are usually classified under three main headings:

Classification of Error



a. Gross error

It mainly covers human mistakes in reading or using instruments and in recording and calculating measurement results. As long as human beings are involved, some gross errors will definitely be committed. Although complete elimination of gross errors is probably impossible, one should try to anticipate and correct them. One common gross error, frequently committed by beginners in measurement work, involves the improper use of instrument.

Elimination

- Errors caused by the loading effect of the voltmeter can be avoided by using it intelligently. For example, a low resistance voltmeter should not be used to measure voltages in a vacuum tube amplifier. In this particular measurement a high-input impedance voltmeter is required.
- Before the measurement is taken, the measuring instrument should be set to zero.
- Great care should be taken in reading and recording the data. Two, three or even more reading should be taken for the quantity under measurement.

b. Systematic errors

These are due to shortcoming of the instruments such as defective or worn parts and effects of the environment on the equipment or the user. This type of error is usually divided into two different categories.

i. Instrumental errors

Instrumental errors are errors inherent in measuring instrument because of their mechanical structure, there are many kinds of instrumental errors depending on the type of instrument used.

Elimination

- Selection of suitable instrument for the particular measurement application.
- Applying correction factors after determining the amount of instrumental error.
- Calibrating the instrument against a standard.

ii. Environmental Errors

Environmental errors are due to conditions external to the measuring device, including .conditions in the area surrounding the instrument such as the effects of changes in temperature, humidity, barometric pressure or of magnetic or electrostatic fields.

Elimination

- By air conditioning, hermetically sealing certain components in the instrument, use of magnetic shields.

c. Random (Residual) Error

These errors are due to unknown causes and occur even when all systematic errors have been accounted for. In well-designed experiment, few random errors usually occur but they become important in high accuracy work.

Elimination

- By increasing the number of readings and using statistical means to obtain the best approximation of true value of the quantity under measurement.

Example 1

A Voltmeter having a sensitivity of 1000 Ω/V , reads 100V units 150 V scale when connected across an unknown resistor in series with a milli-ammeter. When the milli-ammeter reads 5 mA, calculate:

a. The apparent resistance of the unknown resistor.
b. The actual resistance of the unknown resistor.
c. The error due to the loading effect of the voltmeter.

Solution

a. The total circuit resistor equals

$$R_T = \frac{V_T}{I_T} = \frac{100V}{5mA} = 20K\Omega$$

Neglecting the resistance of the milli-ammeter, the value of the unknown resistor is $R_x =$ 20 KΩ

b. The voltmeter is in parallel with the unknown resistor

We have,

$R_T = \frac{R_x R_v}{R_v + R_x}$ or, $R_T R_V + R_T R_X = R_X R_V$

or, unknown resistor value (R_X) = $\frac{R_T R_V}{R_V - R_T} = \frac{20*150}{150-20} = \frac{3000}{130} = 23.077K\Omega$

c. %error = $\frac{actual - apparent\ value}{actaul\ value} * 100\% = \frac{23.077-20}{23.077} * 100\% = 13.33\%$

Example 2

Repeat example 1 of the milliammeter reads 800 mA and the voltmeter reads 40 V on its 150 V scale.

Solution

a. $R_T = \frac{V_T}{I_T} = \frac{40V}{800mA} = \frac{40V}{0.8A} = 50\Omega$

b. $R_V = 1000\ \Omega/\text{V}$ *150 = 150KΩ

$$\therefore R_X = \frac{R_T R_V}{R_V - R_T} = \frac{50 * 150 * 10^3}{150 * 10^3 - 50} = 50.017\Omega$$

c. % error = $\frac{50.017-50}{50.017} * 100\% = 0.034\%$

Statistical Analysis

1. Arithmetic mean

$\bar{X} = \frac{(x_1 + x_2 + x_3 + \cdots \ldots\ldots\ldots + x_n)}{n} = \frac{\sum x}{n}$

Where, $\bar{X}$= arithmetic mean

$x_1, x_2 \ldots \ldots x_n$ = readings taken

$n = number\ of\ readings$

2. Deviation from the mean

$d_1 = x_1 - \bar{X}$

$d_2 = x_2 - \bar{X}$

$d_n = x_n - \bar{X}$

where, d_n is the deviation of the n^th reading from the mean.

3. Average Deviation from the mean

Average Deviation from the mean, D, can be expressed as

$$D = \frac{|d_1| + |d_2| + |d_3| + \cdots \ldots . \ldots + |d_n|}{n} = \frac{\sum |d|}{n}$$

4. Standard Deviation

The standard deviation or the root mean square deviation of a finite number of data is given by (n < 20)

S.D. = S = $\sqrt{\frac{d_1^2+d_2^2+d_3^2+\cdots\ldots+d_n^2}{n-1}} = \sqrt{\frac{\sum d_t^2}{n-1}}$

For infinite number of data (n ≥ 20)

S.D. = σ = $\sqrt{\frac{\sum d_t^2}{n}}$

Note: when the number of observations is greater than 20, S.D is denoted by symbol σ while if the number of observations is less than 20 , the symbol used is S.

5. Variance

The variance is square of standard deviation

i.e. V= $(S.D)^2 = \sigma^2 = \frac{\sum d_t^2}{n}$ (for greater no. of observation greater than 20)

$= S^2 = \frac{\sum d_t^2}{n-1}$ (for no. of observations less than 20)

Example

A circuit was tuned for resonance by eight different student and the value of resonant frequency in KHz were recorded as 532, 548, 543, 535,546, 531, 543 and 536. Calculate

a. The arithmetic mean
b. Deviations from mean
c. The average deviation
d. The standard deviation
e. Variance

Solution:

a. The arithmetic mean of the reading is $\bar{X} = \frac{\sum X}{n}$

$= \frac{532+548+543+535+546+531+543+536}{8}$

= 539.25 KHz.

b. The deviations from mean are

$d_1 = X_1 - \bar{X}$ =532 - 539.25= -7.25 KHz
$d_2 = X_2 - \bar{X}$ =548 - 539.25= +8.75 KHz
$d_3 = X_3 - \bar{X}$ =543 - 539.25= +3.75 KHz

$d_4 = \quad X_4 - \bar{X}$ =535 - 539.25= -4.25 KHz
$d_5 = \quad X_5 - \bar{X}$ =546 - 539.25= +6.75 KHz
$d_6 = \quad X_6 - \bar{X}$ =531 - 539.25= -8.25 KHz
$d_7 = \quad X_7 - \bar{X}$ =543 - 539.25= +3.75 KHz
$d_8 = \quad X_8 - \bar{X}$ =536 - 539.25= -3.25 KHz

c. The average deviation from the mean is
D= $\frac{\sum|d|}{n} = \frac{7.25+8.75+3.75+4.25+6.75+8.25+3.75+3.25}{8} = 5.75\ KHz$

d. The standard deviation is
S.D. = S = $\sqrt{\frac{\Sigma d_t^2}{n-1}}$ since the no. of reading is 8 which is less than 20.
$= \sqrt{\frac{(-7.25)^2+(8.75)^2+(3.75)^2+(-4.25)^2+(6.75)^2+(-8.25)^2+(3.75)^2+(-3.25)^2}{8-1}}$
= 6.54 KHz.

e. Variance (v) = S^2 = 42.77 $(KHz)^2$.

Probability of Errors

Normal Distribution of Errors

Example: Tabulation of voltage Reading

Voltage Reading (Volts)	Number of Readings
99.7	1 i.e. P(99.7)=1/50
99.8	4
99.9	12
100.0	19 Most probable value of true voltage
100.1	10
100.2	3
100.3	1
	50

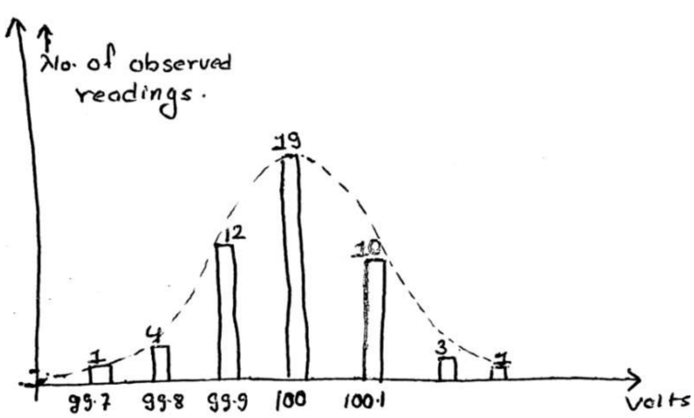


Fig. Histogram showing the frequency of occurrence of the 50 voltage readings. The dashed curve represents the limiting case of the histogram when a large no. of reading as small increments are taken.

The Gaussian or Normal law of errors forms the basis of the analytical study of random effects. The following qualitative statements are based on the normal law:

a. All observations include small disturbing effects called random errors.
b. Random errors can be positive or negative
c. There is an equal probability of positive and negative random errors.

The possibilities as to form the error distribution curve can be stated as :

a. Small errors are more probable than large errors.
b. Large errors are very improbable.
c. There is an equal probability of plus and minus errors so that the probability of a given error will be symmetric about the zero value.

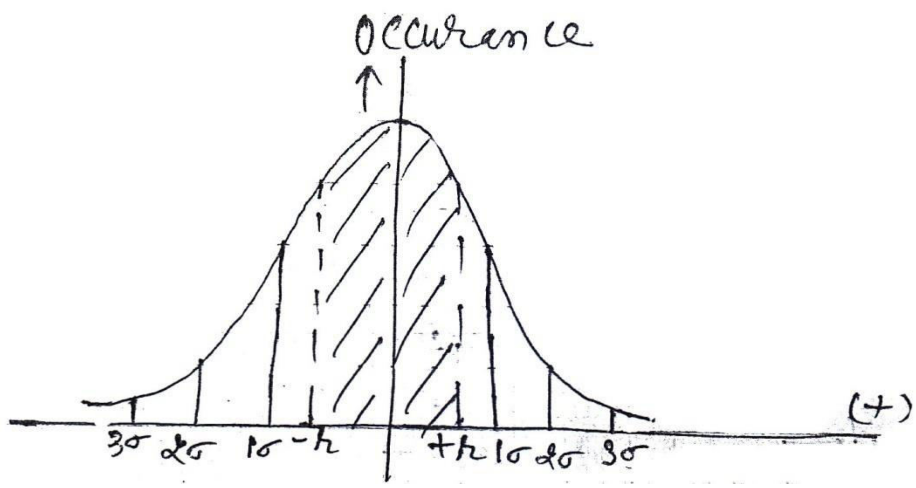


Fig. Curve for the Normal law. The shaded portion indicates the region of probable error where,

r = ± 0.6745σ

Probable error: Probable error, r = ± 0.6745σ

Example 1

The following 10 observations were recorded when measuring a voltage 41.7, 42.0, 41.8, 42.0, 42.1, 41.9, 42.0, 41.9, 42.5 and 41.8 Volt. Find

i. the mean
ii. the standard deviation
iii. the probable error of one reading
iv. the probable error of mean
v. the range

Solution

i. Mean ($\bar{X}$) = $\frac{\sum X}{n} = \frac{419.7}{10} = 41.97\ Volt$

Measured Voltage(X)	Deviation(d)	d^2
41.7	-0.27	0.0729
42.0	+0.03	0.0009
41.8	-0.17	0.0289
42.0	+0.03	0.0009
42.1	+0.13	0.0169
41.9	-0.07	0.0049
42.0	+0.03	0.0009
41.9	-0.07	0.0049
42.5	+0.53	0.2809
41.8	-0.17	0.0289
$\sum X$ = 419.7		$\sum$ d^2 =0.441

ii. Standard Deviation is

S.D =S = $\sqrt{\frac{\Sigma d_t^2}{n-1}}$ since the no. of observation is less than 20

$= \sqrt{\frac{0.441}{10-1}} = 0.22\ Volt.$

iii. Probable error of one reading is

$r_1 = 0.6745S$

= 0.6745*0.22 = 0.15volt.

iv. Probable error of mean is

$$r_m = \frac{r_1}{\sqrt{n-1}} = \frac{0.15}{\sqrt{9}} = 0.05\ Volt$$

v. Range = Max Voltage- Min Voltage = 42.5-41.7 = 0.8 Volt

Note:

i. If n<20 , $r_m = \frac{r_1}{\sqrt{n-1}}$ and

If n>20, $r_m = \frac{r_1}{\sqrt{n}}$

ii. Standard deviation of standard deviation is

$$\sigma_\sigma = \frac{\sigma}{\sqrt{2n}} = \frac{\sigma_m}{\sqrt{n}}$$

iii. Standard deviation of mean is

$\sigma_m = \frac{\sigma}{\sqrt{n}}$ where, n is the number of observations.

Example 2

Following reading were obtained in respect of measurement of capacitor: 1.005, 0.998, 1.001, 1.009, 1.005, 0.991, 0.996, 0.997, 1.008 and 0.994 µF. Calculate

i. Arithmetic mean ii. Deviation from mean iii. Average deviation

iv. Standard deviation v. Variance vi. Range vii. Probable error of one reading

Solution

X	d=X- $\bar{X}$	d^2	|d|
1.005	0.00280	0.00000784	0.00280
0.998	-0.00220	0.00000484	0.00220
1.001	0.00080	0.00000064	0.00080
1.009	0.00880	0.00007744	0.00480
1.005	0.00480	0.00002304	0.00480
0.991	-0.00920	0.00008464	0.00920
0.996	-0.00420	0.00001764	0.00420
0.997	-0.00320	0.00001024	0.00320
1.008	-0.0078	0.00006084	0.00780
0.994	-0.00620	0.00003844	0.00620
ΣX=10.002		Σd^2=0.0003256	Σ|d|=0.0500

i. Arithmetic mean($\bar{X}$) = $\frac{\Sigma x}{n} = \frac{10.002}{10}$ = 1.0002 µF

ii. Average deviation (D) = $\frac{\Sigma|d|}{n} = \frac{0.05}{10}$ = 0.005 µF

iii. Standard deviation(σ)=$\sqrt{\frac{\Sigma d^2}{n-1}} = \sqrt{\frac{0.0003256}{10}}$ = 0.006014797µF

iv. Variance(σ^2) = $(0.006014797)^2$

v. Range = Largest value – smallest value

= 1.009 – 0.991 = 0.018 µF

vi. Probable error of one reading = ±0.6745σ

$r_1 = 0.6745*6*10^{-3}$ µF = $4.047\text{-}10^{-3}$ µF

Limiting Errors

In almost indicating instruments, the accuracy is guaranteed to a certain percentage of full-scale reading. Circuit components such as capacitors, resistors etc. are guaranteed within a certain percentage of their rated value. The limits of these deviations from the specified value are known as limiting errors or guaranteed errors. For example, if the resistance of a resistor is given as 500 Ω ± 10%, the manufacture guarantees that the resistance falls between the limits 450Ω and 550Ω . The manufacturer is not specifying a standard deviations or a probable error but promises that the error is no greater than the limits set.

Example 1

A 0-150 V voltmeter has a guaranteed accuracy of 1% full scale reading. The voltage measured by this instrument is 83 V. Calculate the limiting error in percentage.

Solution

The magnitude of the limiting error is 1% of 150V = $\frac{150}{100} = 1.5V$

∴ % error at a meter indication of 83V is $\frac{1.5}{83} * 100\% = 1.81\%$

Example 2

The voltage generated by a circuit is equally dependent on the value of three resistors and is given by the following equation:

$V_{out} = \frac{R_1 R_2}{R_3}$

If the tolerance of each resistor is 0.1% , what is the maximum error of the generated voltage?

Solution

The highest resulting voltage occurs when R_1 and R_2 are at the maximum value allowed by the tolerance while R_3 is at the lowest value allowed by the tolerance. For a variation of 0.1% the highest value of resistor is 1.001 times the normal value while the lowest value is 0.999 times the normal value.

The resulting value is $V_{out} = \frac{(1.001R_1)(1.001R_2)}{0.999R_3} = 1.003\frac{R_1R_2}{R_3}$

The lowest resulting voltage occurs when the value of R_3 is highest and $R_1 and\ R_2$ are the lowest i.e

$$V_{out} = \frac{(0.999R_1)(0.999R_2)}{1.001R_3} = 0.997\frac{R_1R_2}{R_3}$$

Hence, the total variation of the resulting voltage is ± 0.3% which is the algebraic sum of the individual tolerance.

Example 3

The current passing through a resistor of 100 ± 0.2 Ω is 2.00 ± 0.01 A. Using the relationship P=I^2R , calculate the limiting error in the computed value of power dissipation.

Solution

Expressing the guaranteed limits of both current and resistance in percentages instead of units , we obtain
I=2.00 ± 0.01A = 2.00 ± 0.5%
R= 100 ± 0.2Ω = 100 ± 0.2%
For highest power dissipation, the highest value of resistance and current is used .i.e

P=$(2.01)^2 * 100.2 = 404.82\ W$

Since the power dissipation without tolerance is $P = I^2R = 2^2 \times 100 = 400\ W$, % limiting error of power dissipation is $\frac{400-404.82}{400} * 100\% = -\frac{4.82}{400} * 100\% = -1.205\%$

For the lowest power dissipation, lowest value of resistance and current is used. i.e

P = $(1.99)^2 * 99.8$

= 395.22W

Here % limiting error of power dissipation is $\frac{400-395.22}{400} * 100\%$

$$= \frac{4.78}{400} * 100\% = 1.195 \simeq 1.2\%$$

Hence, the error is ± 1.2% which is two times the 0.5% error of the current plus 0.2% error of the resistor.

Example 4

The resistance of an unknown resistor is determined by the Wheatstone Bridge method. The solution for the unknown resistance is stated as $R_x = \frac{R_1R_2}{R_3}$, where

R_1= 500 Ω ± 1%

R_2= 615 Ω ± 1%

R_3 = 100 Ω ± 0.5%

Calculate

a. The nominal value of the unknown resistor
b. The limiting error in Ω of the unknown resistor and
c. The limiting error in percentage of the unknown resistor

Solution

a. The nominal value of the unknown resistor is $R_x = \frac{R_1R_2}{R_3} = \frac{500*615}{100} = 3075\Omega$
b. The limiting error in ohms of the R_x , the value of R_1 and R_2 must be high and the value of R_3 should be low i.e. $R_x = \frac{(500+5)(615+6.15)}{100-0.5} = \left(\frac{(505)(621.5)}{99.5}\right) = 3152.57\Omega$

   Now, the limiting error is Nominal value minus highest value of unknown resistor i.e.

   Limiting error = (3075-3152.57) Ω

   = -77.57Ω

   Hence, the limiting error of unknown resistor is ± 77.57Ω
c. % limiting error of unknown resistor is $\frac{limiting\ error}{nominal\ value} * 100\%$

   $= \pm\frac{77.57}{3075} * 100\%$

   = ±2.52%

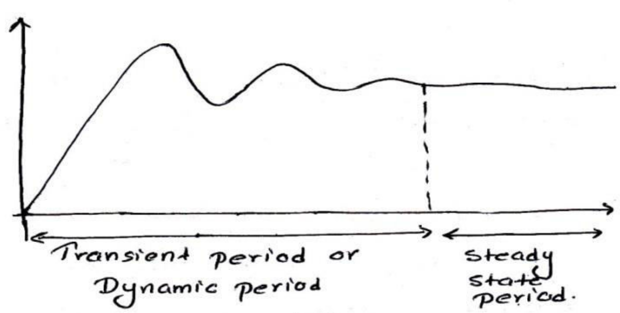
Performance Parameters

It is divided into two categories.

1. Static characteristics
2. Dynamic Characteristics

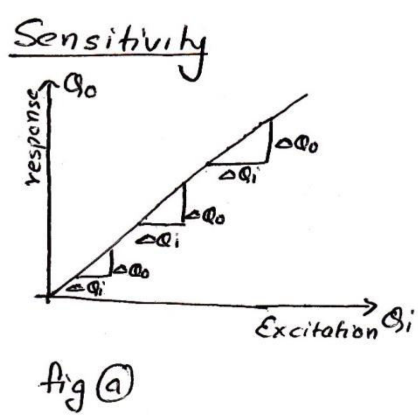
1. Static Characteristics

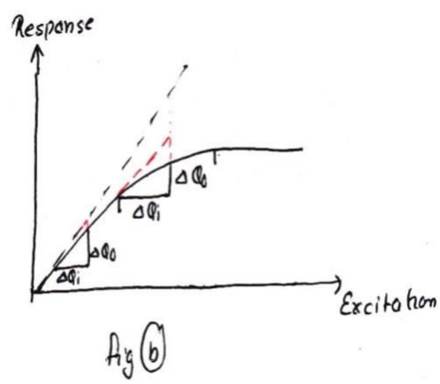
- The static characteristics of a measurement system are those that must be considered when the system or instrument is used to measure a conditions not varying with time. i.e. When steady state condition occurs, some of its static Performance parameters (characteristics) are as follows:



The following are the static characteristics:

a. Accuracy
   It is the closeness with which an instrument reading approaches the true value of the variable being measured.
b. Sensitivity
   It is the ratio of output signal or response of the instrument to a change of input or measured variable.





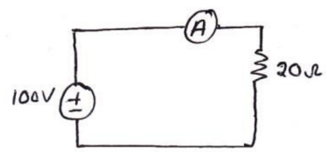
- When the calibration curve is linear as shown in Fig a, then sensitivity is constant throughout the whole range and is defined by the slope of the calibration curve.
- However if the curve is not straight as shown in Fig b, the sensitivity varies with the output.

$$S = \frac{small\ changes\ in\ o/p}{small\ changes\ in\ i/p} = \frac{\Delta Qo}{\Delta Qi} = \frac{mag.of\ o/p}{mag.of\ i/p}$$

c. Precision(Reproducibility)

It refers to the measure of the degree to which successive measurements differ from one another.

To differentiate between accuracy and precision, let us consider an instrument which may be giving reading that is highly repeatable from measurement to measurement, yet far from the true value. Therefore the instrument is highly précised but inaccurate. Therefore precision does not guarantee accuracy.



A1 → 4.5, 4.7, 4.9, 5, 4.6

A2 → 3, 3, 3

d. Drift

Perfect precision means that the instrument has no drift i.e. with a given input, the measured values don't vary with time.

e. Dead zone

It is defined as the largest change of input quantity for which there is no output of the instrument.

f. Resolution

It is the smallest change in measured value to which the instrument will respond.

g. Loading effect

The incapability of the system to faithfully measure, record or control the input signal in undistorted form is called the loading effect.

h. Linearity

- Here, output is linearly proportional to the input. Linear behavior is required for most of the systems as it is desirable. It means constant sensitivity throughout the whole range. It is represented as y = mx + c

  Where, y = output

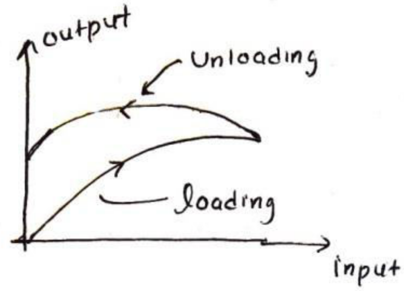
  x = input

  m =slope

  c = intercept

i. Hysteresis

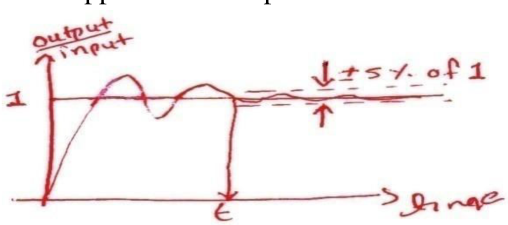
It is a phenomenon which depicts different output effect when loading and unloading whether it is a mechanical or an electrical system. It is non- coincidence of loading and unloading curves as shown in Fig. below.



2. Dynamic Characteristics

Performance criteria based upon dynamic relations constitute the dynamic characteristics. The dynamic characteristics of a measurement system are:

a. Speed of response
   It is defined as the rapidity with which a measurement system responds to changes in the measured quantity.
b. Measuring lag
   It is the delay in the response of a measurement system to changes in the measured quantity.
c. Response Time
   It is defined as the time required by the instrument to settle down to its final steady position after the application of input.



d. Fidelity
   It is defined as the degree to which a measurement system indicates changes in the measured quantity without any dynamic error.
e. Dynamic Error
   It is the difference between the true value of the quantity under measurement changing with time and the value indicated by the measurement system if no static error is assumed.

Classification of Resistances

The classification of resistances from the point of view of measurement is as follows:

i. Low Resistances : $R < 1\Omega$
ii. Medium Resistances: $1\Omega \leq R < 0.1\ M\Omega$
iii. High Resistances: $R \geq 0.1\ M\Omega$

Example

If a voltmeter having full scale voltage of 100V and accuracy of 1% is used to measure i) 80V ii)12V .Calculate the %error in both cases and comment your answer.

i) T.V = 80V

MV = 80V ± 1% of 100V
= 79V or 81V

Therefore, % error = $\frac{TV-MV}{TV}$ * 100%
= $\frac{80-79}{80}$ * 100% = 1.25%

ii) T.V = 12V

MV = 12V±1%of 100V = 13V or 11V

Therefore, % error = $\frac{12-11}{12}$ * 100%
= 8.33%

Comment: we can see that the % error is relatively less when the range of the instrument is closer to the quantity. Thus, selection of range in measurement system plays a great role in minimizing the error.

Bridge Circuit

- It is used to determine unknown electrical parameters such as resistance (R), inductance (L), capacitance (C) and frequency (f).
- It consists of four arms, a detector and power supply.
- Power supply may be AC or DC

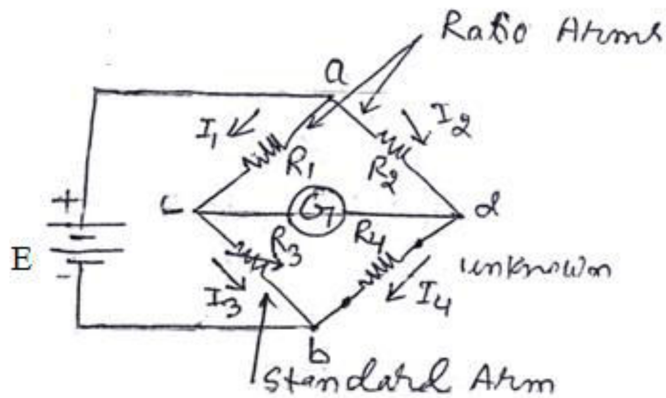
Functions of Detector:

- To detect signal in bridge Circuit.
- Important role to determine balanced condition.
- Headphones, tunable amplifiers, galvanometer are commonly used detector

1. DC Bridge Circuit

a. Wheatstone Bridge

The Wheatstone bridge is an instrument for making comparison measurements and operates upon a null indication principle. It is a very important device used in the measurement of medium resistance due to its accuracy and reliability.



Basic operation

Figure shows the schematic of a Wheatstone bridge. The bridge has four resistive arms together with a source of emf (a battery) and a null detector, usually a galvanometer or other sensitive current meter.

The bridge is said to be balanced when the potential difference across the galvanometer is 0 V so that there is no current through the galvanometer. This condition occurs when the voltage from point c to point a equals the voltage from point d to point a or by referring to the other battery terminal, when the voltage from point c to point b equals the voltage from point d to point b. hence, the bridge is balanced

When $E_{ac} = E_{ad}$

$= I_1R_1 = I_2R_2$ …….(1)

If the galvanometer current is zero , the following conditions also exist:

$I_1 = I_3 = \frac{E}{R_1+R_3}$ …………(2) and

$I_2 = I_4 = \frac{E}{R_2+R_4}$ …………………(3)

Combining equations (1), (2) and (3) and simplifying, we get

$$\frac{R_1}{R_1 + R_3} = \frac{R_2}{R_2 + R_4}$$

$or, R_1R_4 = R_2R_3$ …………………(4)

Equation (4) is the well-known expression for balance of the Wheatstone bridge. If three of the resistance have known values, the fourth may be determined from equation (4) . Hence if R_4 is the unknown resistor, its resistance R_x can be expressed in terms of remaining resistors as

$R_x = \frac{R_2R_3}{R_1}$ …………………….(5)

The measurement of unknown resistance R_x is independent of the characteristics or the calibration of the null-detecting galvanometer provided that the null detector has sufficient sensitivity to indicate the balance position of the bridge with the required degree of precision.

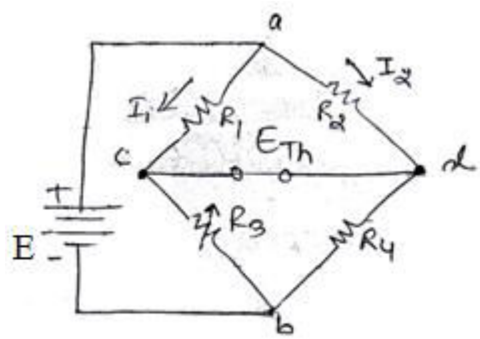
Measurement Errors

The main source of measurement error is found in the limiting errors of the three known resistors. Other errors may include the following:

a. Insufficient sensitivity of the null detector
b. Changes in resistance of the bridge arms due to the heating effect of the current through the resistors
c. Thermal emfs in the bridge circuit or the galvanometer circuit can also cause problems when low value resistors are being measured.
d. Errors due to the resistance of leads and contacts exteriors to the actual bridge circuit play a role in the measurement of very low resistance values. These errors may be reduced by using a Kelvin bridge.

Thevenin Equivalent Circuit

To determine whether or not the galvanometer has the required sensitivity to detect an unbalance condition, it is necessary to calculate the galvanometer current. The Thevenin equivalent circuit is determined by looking into galvanometer terminal c and d. There are two steps to find the Thevenin equivalent circuit.



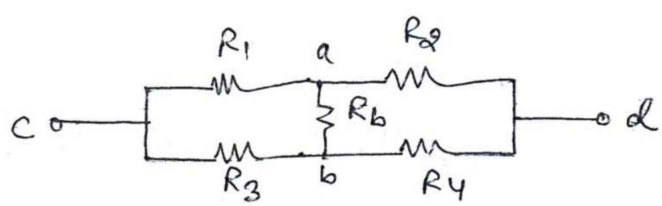
1. Finding the voltage, E_{Th} appearing the terminals c and d when the galvanometer is removed from the circuit. Now, the thevenin or open circuit voltages

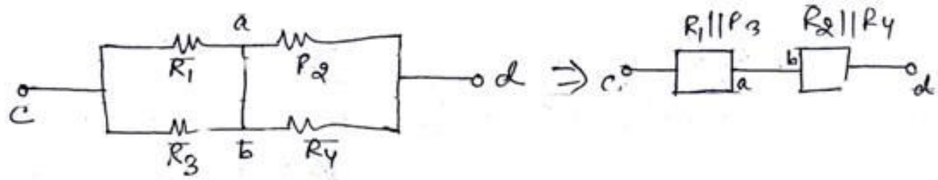
$$E_{Th} = E_{cd} = E_{ac} - E_{ad} = I_1R_1 - I_2R_2$$

$$where, I_1 = \frac{E}{R_1 + R_3} \ \ and\ I_2 = \frac{E}{R_2 + R_4}$$

$$\therefore E_{Th} = E\left(\frac{R_1}{R_1 + R_3} - \frac{R_2}{R_2 + R_4}\right)$$

2. Finding the equivalent resistance looking into the terminals c and d, with the battery replaced by its internal resistance.

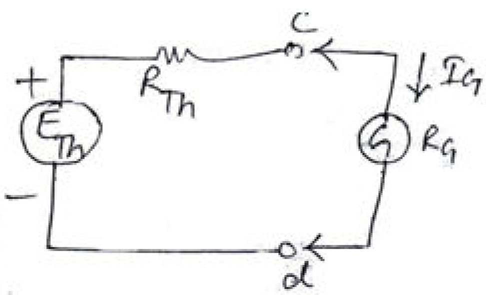




Since, in most cases the internal resistance ,R_b of the battery is extremely low, it can be neglected. i.e $R_b = 0$. So,

$\therefore R_{Th} = (R_1||R_3) + (R_2||R_4) = \left(\frac{R_1 R_3}{R_1+R_3}\right) + \left(\frac{R_2 R_4}{R_2+R_4}\right)$

Hence, the complete Thevenin circuit with the galvanometer connected to terminals c and d is



The galvanometer current ,I_G with its resitance R_G is $I_G = \frac{E_{Th}}{R_{Th}+R_G}$.

Limitations of Wheatstone bridge

i. The use of Wheatstone bridge is limited to the measurement of resistances ranging from a few ohm to several mega ohm.

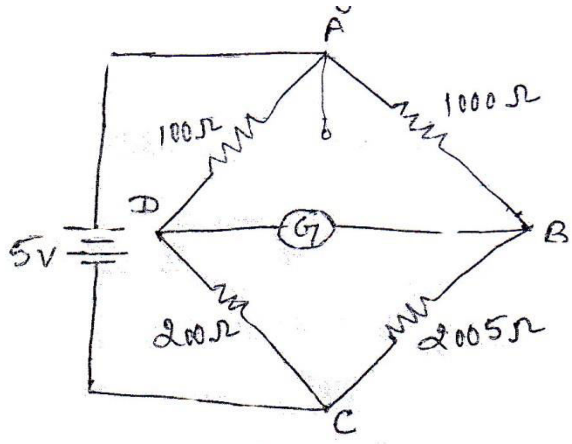
ii. The upper limit is set by the reduction in sensitivity to unbalance caused by high resistance values.

iii. The lower limit for measurement is set by the resistance of the connecting leads and by contact resistance at the binding posts.

Example 1

Figure shows the schematic diagram of a Wheatstone bridge with values of their bridge elements as shown. The battery voltage is 5V and its internal resistance is negligible. The galvanometer has a current sensitivity of 10 mm/μA and an internal resistance of 100Ω calculate the deflection of the galvanometer caused by the 5 Ω unbalance in the arm BC.

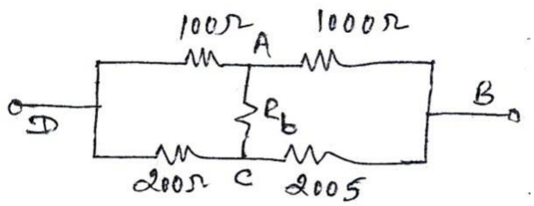
Solution



Since we are interested in finding the current in the galvanometer, the Thevenin equivalent is determined w.r. t. galvanometer terminals B and D.

The potential difference from B to D with the galvanometer removed from the circuit is the Thevenin voltage i.e$E_{Th} = E_{DB} = E\left(\frac{R_1}{R_1+R_3} - \frac{R_2}{R_2+R_4}\right) = 5 * \left(\frac{100}{100+200} - \frac{1000}{1000+2005}\right) \simeq 2.77\ mV$

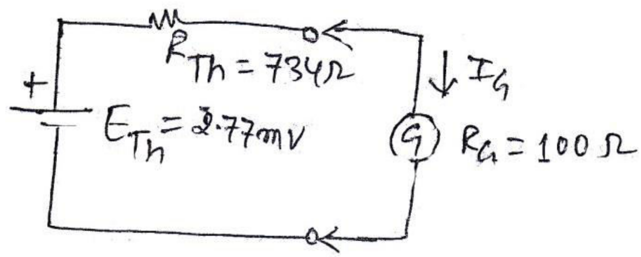
The equivalent Thevenin resistance looking into terminals B and D replacing the battery with its internal resistance is



Since internal resistance of battery, R_b is zero **Ω**, the equivalent resistance is

$$\therefore R_{Th} = (R_1||R_3) + (R_2||R_4) = \left(\frac{100 * 200}{100 + 200} + \frac{1000 * 2005}{1000 + 2005}\right) = 734\ \Omega$$

The Thevenin equivalent circuit is given as,



when the galvanometer is now connected to the output terminals of the equivalent circuit, the current I_G through the galvanometer is

$$I_G = \frac{E_{Th}}{R_{Th} + R_G} = \frac{2.77\ mV}{234\ \Omega + 100\ \Omega} = 3.32\ \mu A$$

Now, the galvanometer deflection is $d = 3.32\mu A * \frac{10mm}{\mu A} = 33.2mm.$

Example 2

The galvanometer of Example 1 is replaced by one with an internal resistance of 500 **Ω** and a current sensitivity of 1 mm/µA. Assuming that a deflection of 1mm can be observed on the galvanometer detecting the 5 **Ω** unbalance in arm BC of the given figure of Example 1.

Solution

Here, $E_{Th} = 2.77mV$

$R_{Th} = 734\Omega$

Now, the new galvanometer is connected to the output terminals resulting in a galvanometer current.

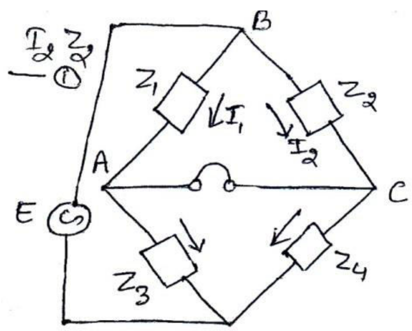
$$I_G = \frac{E_{Th}}{R_{Th} + R_G} = \frac{2.77mV}{734\Omega + 500\Omega} = 2.24\mu A$$

The galvanometer deflection is $d = 2.24\mu A * \frac{1mm}{\mu A} = 2.24\ mm$ which is easily observed.

2. **AC Bridge circuit**

Condition for bridge balance

The ac bridge is a natural outgrowth of the dc bridge and in its basic form consists of four bridge arms, a source of excitation and a null detector. The power source supplies ac voltage to the bridge at the desired frequency. For measurements at low frequencies, the power line may serve as the source of excitation voltage. The null detector must respond to ac unbalance currents and in its cheapest but very effective form consists of a pair of headphones or an ac amplifier.



The general form of an ac bridge is shown in figure. The four bridge arms Z_1, Z_2, Z_3 *and* Z_4 are impedances and the detector is represented by headphone. The balance condition in ac bridge is reached when the detector response is zero or indicates a null. The general equation for bridge balance is obtained by using complex notation for the impedances of the bridge circuit. In complex notation, we can write

$E_{BA} = E_{BC}$ $or,$ $I_1Z_1 = I_2Z_2$…………………(1)

For zero detector current, i.e. the balance condition, the

The currents are

$I_1 = \frac{E}{Z_1+Z_3}$ ……………..(2) and

$I_2 = \frac{E}{Z_2+Z_4}$ ………………(3)

Solving equation (1) ,(2) and (3) and simplifying , we get

$$\frac{Z_1}{Z_1 + Z_3} = \frac{Z_2}{Z_2 + z_4}$$

$\therefore Z_1Z_4 = Z_2Z_3$ …………..(4)

or when using admittances instead of impedances,

$Y_1Y_4 = Y_2Y_3$ …………….(5)

If the impedance is written in the form $Z\angle\theta$, where Z represents the magnitude and θ the phase angle of the complex impedance, equation (4) becomes

$(Z_1\angle\theta_1)(Z_4\angle\theta_4) = (Z_2\angle\theta_2)(Z_3\angle\theta_3)$

$or, Z_1Z_4\angle(\theta_1 + \theta_4) = Z_2Z_3\angle(\theta_2 + \theta_3)$ ………………(6)

Equation (6) shows that following two conditions must be met simultaneously when balancing an ac bridge i.e.

i. The magnitude of the impedance satisfy the relationship

$$Z_1Z_4 = Z_2Z_3$$

Or the products of the magnitude of the opposite arms must be equal.

ii. The sum of the phase angles of the opposite arms must be equal i.e.

$$\angle\theta_1 + \angle\theta_4 = \angle\theta_2 + \angle\theta_3$$

Inductance Bridge

a. Maxwell Bridge
b. Hay Bridge

a. Maxwell Bridge

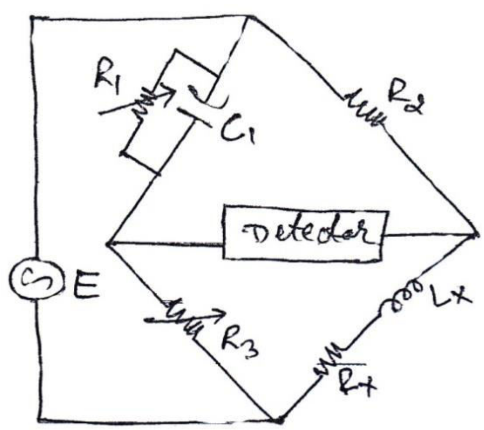
The Maxwell Bridge measures an unknown inductance in terms of a known capacitance. The general equation for balance of the ac bridge is

$Z_1 Z_x = Z_2 Z_3$

$Z_x = \frac{Z_2 Z_3}{Z_1} = Z_2 Z_3 Y_1$

where $Z_2 = R_2; Z_3 = R_3; Y_1 = \frac{1}{R_1} + jwc_1$

So, $Z_x = R_x + jwL_x = R_2 R_3 \left(\frac{1}{R_1} + jwc_1\right)$



Separation of the real and imaginary terms yields

$R_x = \frac{R_2 R_3}{R_1}$ …………………..(1)

$L_x = R_2 R_3 C_1$ …………….(2)

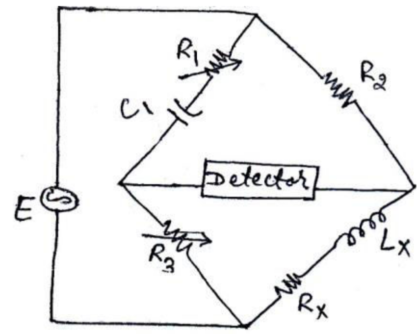
Note : the expression for Q-factor is $Q = \frac{wL}{R}$.

Limitations of Maxwell Bridge

1. It is limited to the measurement of medium- Q coils (1<Q<10). It is unsuited for the measurement of coils with a very low Q-value (Q<1) because of balance convergence problems.
2. This bridge requires a variable standard capacitor which may be very expensive if calibrated to a high degree of accuracy.

b. Hay Bridge

The Hay bridge is suited for the measurement of high Q inductors especially for those inductors having a Q greater than 10.



The general equation for ac bridge balance is

$$Z_1 Z_x = Z_2 Z_3$$
$where,$
$$Z_1 = R_1 - \frac{j}{wc_1}; \quad Z_2 = R_2$$
$$Z_3 = R_3; \quad Z_x = R_x + jwL_x$$
So, $\left(R_1 - \frac{j}{wC_1}\right)(R_x + jwL_x) = R_2 R_3$

Separating the real and imaginary terms,

$R_1 R_x + \frac{L_x}{C_1} = R_2 R_3$ …………………..(1)

$$-\frac{R_x}{wc_1} + wL_x R_1 = 0$$

$or, \frac{R_x}{wC_1} = wL_x R_1$ …………………….. (2)

$solving\ equation\ (1) and\ (2)$, we get

$$R_x = \frac{w^2 C_1^2 R_1 R_2 R_3}{1 + w^2 C_1^2 R_1^2}$$
and
$$L_x = \frac{R_2 R_3 C_1}{1 + w^2 C_1^2 R_1^2}$$

Also, the sum of the opposite sets of phase angles must be equal i.e. the inductive phase angle must be equal to the capacitive phase angle since the resistive angles are zero.

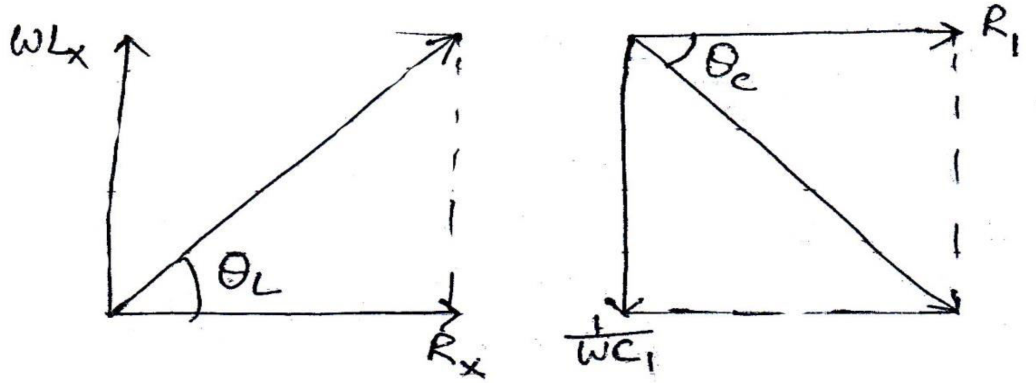


Fig. Impedance triangles illustrate inductive and capacitive phase angles.

Now, the tangent of the inductive phase angle equals

$tan\theta_L = \frac{X_L}{R} = \frac{wL_X}{R_x} = Q = quality\ of\ coil$

and that of capacitive is

$tan\theta_C = \frac{X_C}{R} = 1/wC_1R_1$

When two phase angles are equal, their tangents are also equal,

$tan\theta_L = tan\theta_c$

$or, Q = 1/wC_1R_1$

And hence, $L_x = \frac{R_2R_3C_1}{1+\left(\frac{1}{Q}\right)^2}$

For a value of Q greater than 10, the term $\left(\frac{1}{Q}\right)^2$ can be neglected.

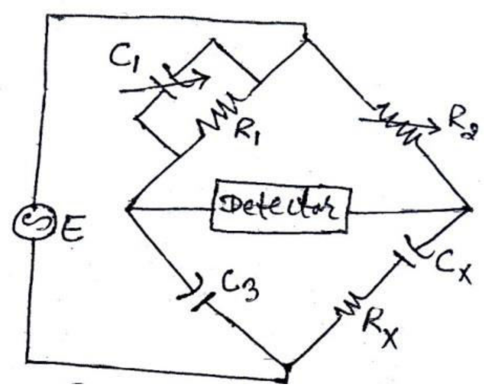
So, $L_x = R_2R_3C_1$

For a value of Q smaller than 10, the term $\left(\frac{1}{Q}\right)^2$ becomes important and cannot be neglected. So, the Maxwell bridge is more suitable in this case.

Capacitance Bridge (Schering Bridge)

It is one of the most important ac bridges extensively used for the measurement of capacitors.
The general equation for ac bridge balance is

$Z_1Z_x = Z_2Z_3$ or, $Z_x = Z_2Z_3Y_1$



where,

$Z_2 = R_2;\ Z_3 = -j/wc_3$

$Y_1 = \frac{1}{R_1} + jwc_1$ and $Z_x = R_x - j/wc_x$

$or, Z_x = R_x - \frac{j}{wC_x} = R_2\left(-\frac{j}{wC_3}\right)\left(\frac{1}{R_1} + jwc_1\right) = \frac{R_2C_1}{C_3} - \frac{jR_2}{wC_3R_1}$

Equating the real and imaginary terms,

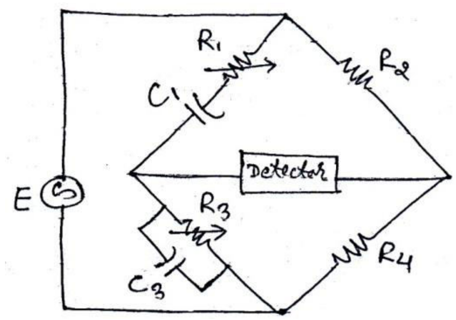
$R_x = \frac{R_2C_1}{C_3} = R_2\frac{C_1}{C_3}\ \ and\ C_x = \ c_3\frac{R_1}{R_2}$

Wien Bridge

It is an ac bridge used to measure the frequency. A Wien Bridge is used as a notch filter in the harmonic distortion analyzer. The Wien bridge also finds application in audio and HF oscillators as the frequency determining element. The general equation for the ac bridge is,

$Z_1Z_4 = Z_2Z_3$

Or, $Z_2 = Z_1 Z_4 Y_3$
where,
$Z_2 = R_2;\;\; Z_4 = R_4$
$Z_1 = R_1 - \frac{j}{wC_1}$ and $Y_3 = \frac{1}{R_3} + jwc_3$



Now, $Z_2 = R_2 = \left(R_1 - \frac{j}{wC_1}\right)(R_4)\left(\frac{1}{R_3} + jwc_3\right)$
$= \frac{R_1R_4}{R_3} + \frac{R_4C_3}{C_1} + jwc_3R_1R_4 - \frac{jR_4}{wC_1R_3}$
Equating real and imaginary terms,
$$R_2 = \frac{R_1R_4}{R_3} + \frac{R_4C_3}{C_1}$$
which reduces to
$\frac{R_2}{R_4} = \frac{R_1}{R_3} + \frac{C_3}{C_4}$ ……………(1)
and $wC_3R_1R_4 - \frac{R_4}{wC_1R_3} = 0$
Or, $wC_3R_1R_4 = R_4/wC_1R_3$
Or, $w^2 = \frac{1}{C_1C_3R_1R_3}$ $where,\; w = 2\pi f\; and\; solving\; for\; f\;, we\; get$
$w = \frac{1}{\sqrt{C_1C_3R_1R_3}}$
Or, $f = \frac{1}{2\pi\sqrt{C_1C_3R_1R_3}}$……………….(2)
In most Wien Bridge circuits, the components are chosen such that $R_1 = R_3$ and $C_1 = C_3$. So, equation (1) and (2) becomes
$\frac{R_2}{R_4} = 2$ and
$f = \frac{1}{2\pi RC}$ is the general expression for the frequency of the Wien bridge.

Application of Wien Bridge
- It is primarily used in a.c bridges to measure frequency
- It may be employed in harmonic distortion analyzer where it is used as a notch filter
- It is used in audio and HF oscillators as the frequency determining device.

Example 1

The impedance of the basic ac bridge are given as follows:
$Z_1 = 100\Omega\angle 80°(inductive\ impedance)$
$Z_2 = 250\Omega(pure\ resistance)$
$Z_3 = 400\Omega\angle 30°(inductive\ impedance)$
$Z_4 = unknown$; Determine the constants of unknown arm.

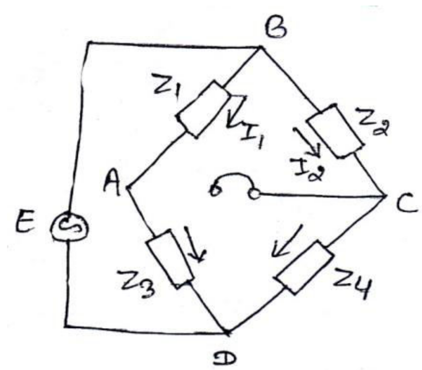
Solution

$$Z_1Z_4 = Z_2Z_3$$
$$or, |Z_1|Z_4\angle\theta_1 = |Z_2||Z_3|\angle\theta_2 + \theta_3$$
$$\therefore Z_4 = \frac{|Z_2||Z_3|}{|Z_1|}\frac{\angle\theta_2 + \theta_3}{\angle\theta_1} = \frac{|Z_2||Z_3|}{|Z_1|}\angle\theta_2 + \theta_2 - \theta_1$$
$$= \frac{250*400}{100}\angle(0° + 30° - 80°)$$
$= 1{,}000\Omega\angle - 50°$ indicates that the element is capacitor.

Example 2

The ac bridge is in balance with the following constants: arm AB, $R = 450\ \Omega$; arm BC, R=300 Ω in series with C=0.265µF; arm CD = unknown; arm DA, R=200 Ω in series with L=15.9 mH. The oscillator frequency is 1 KHz . Find the constants of arm CD.

Solution



The general equation for ac bridge balance is
$Z_1Z_4 = Z_2Z_3$
where,
$Z_1 = R = 450\ \Omega$
$$Z_2 = R - \frac{j}{wC} = (300 - j600)\ \Omega$$
$Z_3 = R + jwL = (200 + j100)\ \Omega$
$Z_4 = unknown$
Now, $Z_4 = \frac{Z_2Z_3}{Z_1} = \frac{(300-j600)(200+j100)}{450}$
$$= \frac{60000 + j30000 - j120000 + 60000}{450}$$
$$= \frac{120000 - j90000}{450}$$
$$= (266.67 - j200)\Omega$$
$$= \left(R_4 - \frac{j}{wC_4}\right)\Omega$$

This result indicates that Z_4 has a resistor 266.67 Ω in series with capacitor at a frequency of 1KHz. Since capacitive reactance

$X_{C_4} = \frac{1}{2\pi f C_4}$

or, $\frac{1}{2\pi f C_4} = 200$

or, $C_4 = 1/(2\pi * 1000 * 200)$

$= \frac{1}{1256637.06}$

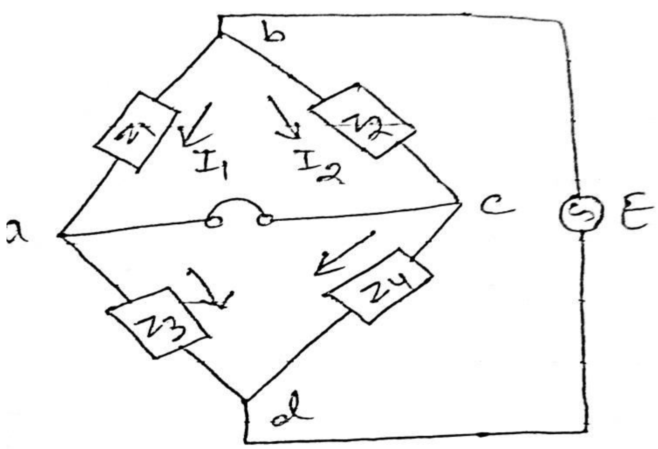
$= 0.7958\ \mu F.$

and $R_4 = 266.67\ \Omega$

Example 3

An ac bridge circuit working at 1000 Hz is shown. Arm ab is a 0.2 μF pure capacitor; arm bc is a 500 Ω pure resistance; arm cd contains an unknown impedance and arm da has a 300 Ω resistance in parallel with 0.1 μF capacitor. Find the R and C or L contains of arm cd considering it as a series circuit.

Solution



Impedance of arm ab is $Z_1 = \frac{1}{2\pi f C_1} = \frac{1}{2\pi * 1000 * 0.2 * 10^{-6}} = 795.77\ \Omega$

$\therefore Z_1 = 795.77\Omega\angle - 90°\ since\ it\ is\ a\ pure\ capacitance.$

Impedance of arm bc is $Z_2 = 500\Omega$

$\therefore Z_2 = 500\Omega\angle 0°\ since\ it\ is\ a\ pure\ resistance$

Impedance of arm cd is $Z_4 = unknown$

and impedance of arm da is $Z_3 = R_3 || X_{C_3}$

$or, Y_3 = \frac{1}{R_3} + jwX_3 = \frac{1}{300} + j2\pi * 1000 * 0.1 * 10^{-6}$

$= 3.33 * 10^{-3} + j6.28 * 10^{-4}$

$= 3.39 * 10^{-3}\angle - 10.68°$

$\therefore Z_3 = \frac{1}{Y_3} = 294.99\angle - 10.68°$

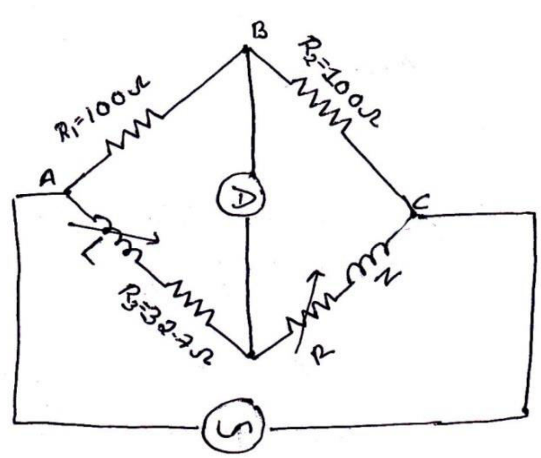
Now, for ac bridge balance

$$Z_1Z_4 = Z_2Z_3$$
$$Z_4 = \frac{Z_2Z_3}{Z_1} = \frac{500\llcorner 0° * 294.99}{795.77\angle -90°} = 185.35\angle 0° - 10.68° + 90°$$
$$\therefore Z_4 = 185.35\angle 79.32°$$
The positive angle for impedance indicates that the branch consists of a series R-L circuit
For resistance, $R_4 = |Z_4|cos\theta_4$
=185.35 cos79.32° = 34.35Ω
For inductive reactance, $X_{L_4} = |Z_4|sin\theta_4$
= 185.35sin79.32° = 182.14
So, $X_{L_4} = wL_4$
$$L_4 = \frac{182.14}{w} = \frac{182.14}{2\pi * 1000} = 28.99\, mH \;\simeq 29\, mH$$

Example 4

Four arms of the bridge are as follows. AB and BC are non-reactive Resistors of 100Ω each , DA is standard variable inductor L in series with a resistance 32.7Ω and CD comprises a standard variable resistor R. in series with the coil of unknown impedance . Balance is obtained when L= 47.8 mH and R= 1.36 Ω .find the resistance and reactance of the coil.

Solution



$z_1 = 100\Omega = R_1$
$z_2 = 100\Omega \; = R_2$
$z_3 = R_3 + jwL_3 = 32.7 + jw * 47.8$
$z_4 = R_4 + R + jwL_4$

For balanced condition,

$Z_1Z_4 = Z_2Z_3$
$R1 * (R_4 + R + jwL_4) = R_2 * \left(R_3 + jw\, L_3\right)$
$R_1R_4 + RR_1 + jwL_4R_i = R_2R_3 + jwL.R_2$

Equating real parts
$R_1R_4 + RR_1 = R_2R_3$
$R_4 = \frac{R_2R_3 - R_1R}{R_1}$
$R_4 = \frac{100 * 32.7 + 100 - 1.36}{100}$
$R_4 = 31.34\Omega$

Equating the imaginary parts,
$L_4R_1 = LR_2$
or, $L_4 = \frac{LR_2}{R_1}$ = 47.8*100/100 = 47.8 mH

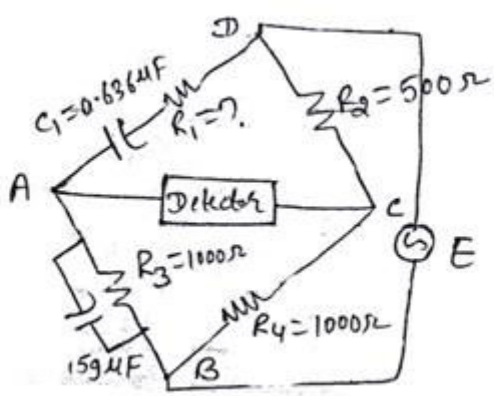
Example 5

An ac bridge has the following constants arm AB, R= 1000 Ω in parallel with C= 0.159 μF; BC, R=1000 Ω; CD, R=500 Ω; DA, C=0.636 μF in series with an unknown resistance. Find the frequency for which this bridge is in balance and determine the value of the resistance in arm DA to produce this balance.

Solution

If we draw the sketch of the given a.c. bridge, we find it is Wien Bridge.

So, the frequency for Wien Bridge is $f = \frac{1}{2\pi\sqrt{C_1C_3R_1R_2}}$..........(1)

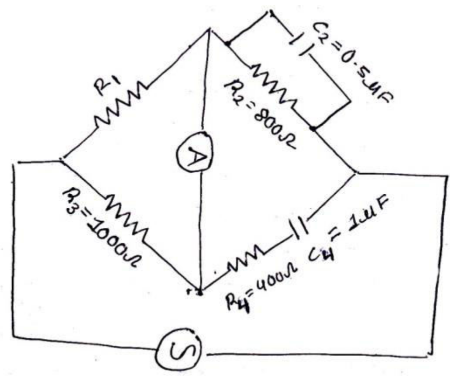


and
$\frac{R_2}{R_4} = \frac{R_1}{R_3} + \frac{C_3}{C_1}$
$\frac{500}{1000} = \frac{R_1}{1000} + \frac{0.159 * 10^{-6}}{0.636 * 10^{-6}}$
$or, 0.5 = \frac{R_1}{1000} + 0.25$
$R_1 = 0.25 * 1000 = 250\,\Omega$
and f = $\frac{1}{\sqrt{(2\pi\sqrt{0.63} * 10^{-6} * 0.159 * 10^{-6} * 250 * 1000)}}$= 1000.97Hz.

Example 6

An ac bridge with terminals A,B,C,D (consecutively marked) has in arm AB a pure resistance ; arm BC a resistance of 800 Ω in parallel with a capacitor of 0.5μF; arm CD a resistance of 400 Ω in series with a capacitor 1μF; arm DA a resistance of 1000 Ω. Obtain the value of frequency for which the bridge can be balanced by first deriving the balance equations. Also, Calculate the value of resistance in arm AB to produce balance.

Solution



$$Z_1 = R_1$$
$$Y_2 = \frac{1}{R^2} + \frac{1}{-j/wc_2} = \frac{1}{R^2} + jwc_2$$
$$Z_3 = R_3\ , Z_4 = R_4 - \frac{j}{wc_4}$$
$$Under\ the\ Balanced\ condition$$
$$Z_1Z_4 = Z_2Z_3$$
$$R_1 * \left(R_4 - \frac{j}{wC_4}\right) = R_3 * 1/(\frac{1}{R_2} + jwC_2)$$
$$R_1 * \left(R_4 - \frac{j}{wC_4}\right) * \left(\frac{1}{R_2} + jwC_2\right) = R_3$$
$$\frac{R_4}{R_2} + \frac{C_2}{C_4} + j\left(wc_2R_4 - \frac{1}{wR_2C_4}\right) = \frac{R_3}{R_1}$$

$$Equating\ real\ parts$$
$$\frac{R_4}{R_2} + \frac{C_2}{C_4} = \frac{R_3}{R_1}$$
$$R_1 = \frac{R_3}{\frac{R_4}{R_2} + \frac{C_2}{C_4}} = \frac{1000}{\frac{400}{800} + \frac{0.5}{1}} = 1000\Omega$$

Equating imaginary parts

$$wC_2R_4 - \frac{1}{WR_2C_4} = 0$$

$$wC_2R_4 = \frac{1}{wR_2C_4}$$

$$w^2 = \frac{1}{C_2C_4R_2R_4}$$

$$f = \frac{1}{2\pi\sqrt{R_2R_4C_2C_4}}$$

$$\therefore f = \frac{1}{2\pi\sqrt{800 * 400 * 0.5 * 1 * 10^{-12}}} = 397.88Hz$$